# Suborno Deb Bappon

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## Education

**University of Saskatchewan** **Sep. 2023 – Aug. 2025 (Tentative)**
*Master of Science in Computer Science – Grade: **89.5%*** *Saskatoon, Saskatchewan*

**Chittagong University of Engineering & Technology** **Feb. 2017 – Sep. 2022**
*Bachelor of Science in Computer Science & Engineering – CGPA: **3.76/4.00 (Merit: 3rd among 128)*** *Chattogram, Bangladesh*

## Skills

**Programming Languages**: Python, C, C++, JavaScript, HTML, CSS
**AI / Machine Learning**: scikit-learn, PyTorch, TensorFlow, LangChain, OpenAI Agents SDK, CrewAI
**Data Analysis & Visualization**: Power BI, Excel, NumPy, Pandas, Matplotlib, Seaborn
**Databases**: PostgreSQL, MySQL, SQLite
**Web Development**: Flask, Django, Gradio
**Cloud Platforms**: Azure
**Version Control & Collaboration**: Git, GitHub
**Tools & Environments**: VS Code, JupyterLab, Spyder, Cursor, Windows, Linux, macOS
**Testing & Practices**: Unit, Integration & Regression Testing; TDD; Agile (Scrum); Performance Optimization; Debugging

## Experience

**Extreme Solutions Ltd.** **Jan. 2022 – Apr. 2022**
*Software Developer Intern* *Chattogram, Bangladesh*

- Optimized backend applications in Python/Flask within Agile sprints, reducing API response times by 35% and improving scalability for analytics tools.
- Designed and deployed predictive analytics pipelines, increasing forecast accuracy by 20% and saving 50+ hours per month in manual analysis.
- Collaborated via Git with cross-functional teams to deliver production-ready software ahead of schedule, boosting client satisfaction by 15%.

**University of Saskatchewan** **Sep. 2023 – Aug. 2025 (Tentative)**
*Graduate Research Assistant – Software Research Lab* *Saskatoon, Saskatchewan*

- Built scalable Python data pipelines to extract, clean, and analyze datasets from Stack Overflow and GitHub, improving processing efficiency by 40% and enabling deeper research insights.
- Deployed LLM-powered tools for code comment generation and Stack Overflow answer enhancement, improving comment quality by 30% and achieving 84.5% adoption among developers.
- Developed multi-agent AI systems to accelerate software prototyping and documentation, cutting development time by 60% and reducing manual documentation work by 80%.

**University of Saskatchewan** **Sep. 2023 – Aug. 2025 (Tentative)**
*Graduate Teaching Assistant – Dept. of Computer Science* *Saskatoon, Saskatchewan*

- Guided students through data-driven programming projects, enhancing problem-solving skills and improving project completion rates by 20%.
- Evaluated assignments using structured rubrics to identify learning gaps, contributing to a 15% improvement in final exam scores across 30+ students.
- Mentored students on documentation, reporting, and code quality, increasing industry-ready project submissions by 35%.

**Eastern University** **Feb. 2023 – Jul. 2023**
*Lecturer – Dept. of Computer Science & Engineering* *Dhaka, Bangladesh*

- Delivered lectures and labs on programming, databases, and machine learning with real-world applications, increasing student lab performance by 22%.
- Mentored undergraduates on coding practices and project delivery, raising successful project completion rates by 30%.
- Redesigned curriculum to align 80% of content with emerging technologies, improving student internship placement success.

**Extreme Solutions Ltd.** **Sep. 2022 – Feb. 2023**
*Data Analyst* *Chattogram, Bangladesh*

- Built interactive Power BI dashboards and automated SQL/DAX reports, cutting manual reporting time by 45% and enabling faster decision-making for leadership.
- Cleaned, transformed, and validated large datasets using Python (Pandas, NumPy) and Power Query, improving data accuracy for business reports by 30%.
- Standardized data governance processes in collaboration with IT and business teams, reducing inconsistencies by 25% and ensuring compliance with corporate policies.

## Selected Projects

**Neora: Multi-Agent System for Rapid Software Prototyping** –*Tech Stack: Python, CrewAI, Gemini, Gradio, Docker* **[GitHub]**

- Built an autonomous multi-agent system converting plain-text requirements into Python modules and UI prototypes, reducing MVP development time by 60%.
- Designed modular YAML pipelines for adaptable role-specific LLM reasoning across models like Gemini 2.5 Flash.
- Auto-generated full software architectures and unit-testable implementations, cutting manual design effort by 80%.

**Repolish: Automated ReadME File Generator** – *Tech Stack: Python, CrewAI, OpenAI, Gemini, Gradio, CSS* **[GitHub]**

- Created an AI-driven tool to parse repositories and generate professional README files, reducing documentation time by 80%.
- Developed a Gradio-based UI with integrated LLM agents, cutting contributor onboarding time by 60%.
- Automated GitHub PR workflows to update documentation, reducing release cycles from hours to minutes.

**AUTOGENICS** – *Tech Stack: Python, Flask, LangChain, Gemini/GPT API, HTML/CSS, SQL* **[GitHub]**

- Designed an LLM pipeline that generates context-aware inline comments for Stack Overflow answers, improving comment quality by 30%.
- Implemented a context extraction pipeline from question text and a noise filtration mechanism to eliminate irrelevant comments, producing concise, high-value annotations.
- Created a browser-integrated prototype leveraging DOM manipulation with Flask APIs for seamless, real-time user interaction.

**Farmona – Smart Crop Prediction & Agronomic Analysis** – *Python, Scikit-learn, Pandas, NumPy, Streamlit* **[GitHub]**

- Developed an end-to-end ML pipeline with feature engineering of agronomic indices, leakage-safe preprocessing, and automated model selection via GridSearchCV.
- Benchmarked multiple classifiers (LR, RF, Gradient Boosting, SVC, KNN, XGBoost), achieving high-accuracy of around 99% in crop prediction with explainability through permutation importance and diagnostic plots.
- Built and deployed an interactive Streamlit app enabling real-time predictions, and surfacing class probabilities—packaged with reproducible deployment artifacts.

**Power BI Data Jobs Dashboard** – *Power BI, DAX, Power Query, ETL, SQL, Data Modelling, UX Design* **[GitHub]**

- Created an interactive dashboard analyzing 479K+ data job postings to identify top in-demand skills, salary ranges, and hiring trends—supporting HR teams and job seekers in career planning and recruitment strategy.
- Designed a star schema data model and optimized DAX measures, improving query speed and drill-down usability by 35%.
- Delivered an executive-ready summary view that was adopted by 3+ HR teams as a decision-support tool for workforce planning.

**Blinkit Sales Analysis Dashboard** – *Power BI, DAX, Power Query, Excel* **[GitHub]**

- Developed a Power BI dashboard revealing sales performance trends, customer rating patterns, and outlet profitability, enabling retail managers to refine product placement and pricing strategies.
- Cleaned and modeled raw retail data in Excel and Power BI, using DAX measures and visualizations to highlight product demand by fat content, region, and outlet size.
- Identified profitable Tier 3, medium-sized outlets and growing preference for low-fat products, influencing stocking and promotional decisions.

## Publications

1. **Suborno Deb Bappon**, Saikat Mondal, and Banani Roy. "AUTOGENICS: Automated Generation of Context-Aware Inline Comments for Code Snippets on Programming Q&A Sites Using LLM." In *2024 IEEE International Conference on Source Code Analysis and Manipulation (SCAM)* (pp. 24–35). IEEE.
2. Saikat Mondal, **Suborno Deb Bappon**, and Chanchal K. Roy. "Enhancing User Interaction in ChatGPT: Characterizing and Consolidating Multiple Prompts for Issue Resolution." In *2024 IEEE/ACM 21st International Conference on Mining Software Repositories (MSR)* (pp. 222–226). IEEE.
3. **Suborno Deb Bappon**, Ashim Dey, Shahriar Mahmud Sabuj, and Annesha Das. "Toward a Machine Learning Approach to Predict the $CO_2$ Rating of Fuel-Consuming Vehicles in Canada." In *2022 25th International Conference on Computer and Information Technology (ICCIT)* (pp. 384–389). IEEE.

## Certifications

Machine Learning with Python (IBM)
Neural Networks & Deep Learning (DeepLearning.AI)
Natural Language Processing with TensorFlow (DeepLearning.AI)
Problem Solving (Intermediate) (HackerRank)

## Leadership / Extracurricular

**Vice President Internal,** Computer Science Graduate Council, University of Saskatchewan (Sep. 2024 – Present)
**Mentor,** Eastern University Computing Club (May 2023 – Jul. 2023)

## Honors & Awards

**Best Research Award,** CMPT 854: Human-Centric Software Renovation, University of Saskatchewan (Spring 2024)
**75th Anniversary Recruitment Scholarship,** CGPS, University of Saskatchewan (2023 – 2025)
**Dean's Award,** Faculty of Electrical & Computer Engineering, Chittagong University of Engineering & Technology (Sep. 2022)

## References

Available upon request.